
ICT Study Guide:

Part 1: Understanding Number Systems

A Number System is a way to represent values. In ICT, we use four primary systems depending on how data is being processed.

1. The Four Main Types

- **Binary (Base 2):** Uses only two digits: 0 and 1. This is the "language" of computers.
- **Decimal (Base 10):** Uses digits 0 to 9. This is the standard system humans use for daily counting.
- **Octal (Base 8):** Uses digits 0 to 7.
- **Hexadecimal (Base 16):** Uses digits 0 to 9 and letters A to F.

2. Identifying the Base

You can tell which system a number belongs to by looking at its subscript:

- (1001)₂ -> Binary
- (1001)₈ -> Octal
- (1001)₁₀ -> Decimal

Part 2: The Hexadecimal Map

Since Hexadecimal is Base 16, we run out of single digits after 9. To represent values 10 through 15, we use letters:

Decimal	Hexadecimal
0 - 9	0 - 9
10	A

11	B
12	C
13	D
14	E
15	F

Part 3: Converting Decimal to Binary

To convert a Decimal number (like 57) into Binary, we use the **Repeated Division Method**.

Step-by-Step Process:

1. Divide the decimal number by 2.
2. Write down the remainder (it will always be 0 or 1) to the side.
3. Divide the result again by 2 and continue until you reach 0.
4. **Crucial Step:** Read the remainders from the bottom to the top.

Example: Converting 57 to Binary

- $57 / 2 = 28$ remainder **1** (LSB)
- $28 / 2 = 14$ remainder **0**
- $14 / 2 = 7$ remainder **0**
- $7 / 2 = 3$ remainder **1**
- $3 / 2 = 1$ remainder **1**
- $1 / 2 = 0$ remainder **1** (MSB)

Result: (111001)₂

Part 4: Key Terminology

When you look at your final binary string, the position of the bits (digits) matters:

- **MSB (Most Significant Bit):** The first bit on the left (the last remainder you calculated). It holds the highest value.
- **LSB (Least Significant Bit):** The last bit on the right (the first remainder you calculated). It holds the lowest value.

Study Tip:

Always double-check your division. Even one wrong remainder will change the entire binary value! To check your work, try adding the powers of 2 for every "1" in your answer:

$$\$32 + 16 + 8 + 1 = 57\$.$$

This layout is ready for a quick copy-paste into Word. I've kept the formatting standard and avoided complex symbols so it stays clean.

Part 5: Converting Decimal to Octal

To convert a Decimal number to Octal (Base 8), we use the repeated division method, but we divide by **8** instead of 2.

Example: Converting 746 to Octal

- $746 / 8 = 93$ remainder **2** (LSB)
- $93 / 8 = 11$ remainder **5**
- $11 / 8 = 1$ remainder **3**
- $1 / 8 = 0$ remainder **1** (MSB)

Result: (1352)₈

Part 6: Converting Binary to Decimal

To convert from Binary back to Decimal, we use **Positional Notation**. This means multiplying each bit by the power of 2 based on its position (starting from 0 on the far right).

Example: Converting (101101)₂ to Decimal

$$(1 \times 2^5) + (0 \times 2^4) + (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) = 45$$

- $32 + 0 + 8 + 4 + 0 + 1 = 45$

Result: $(45)_{10}$